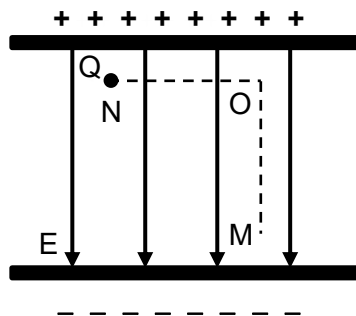


21. The figure below shows a charged particle Q in a uniform electric field E.



Which of the following statements about Q is correct?

- A** No work is done when the positive charge moves from N to O then to M.
- B** No work is done when the positive charge moves from O to M only.
- C** Work is done by field when the positive charge moves from M to O, but work is done by external force when it moves from O to M.
- D** Work done on the positive charge when it moves from N to O to M is the same as when it moves from N to M directly.

Answer: D

Equipotential lines uniform electric field is perpendicular to the electric field lines, so

- no work done moving a charge from N to O or vice versa
- work is done moving a charge from O to M or vice versa

Potential at top > potential at bottom

- When positive charge moves from O to M, change in potential $\Delta V = V_M - V_O < 0$, so change in p.e. $= Q\Delta V < 0$. Decrease in p.e. indicate negative work done by external force or positive work done by the electric force (or field)
- When positive charge moves from M to O, work done by the external force

Work done on positive charge is independent of path taken but initial and final position.